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Safety data sheet

BASF safety data sheet. This is a translation of the country-specific safety data sheet into a language other than that required by law. This document does not replace the safety data sheet provided according to Regulation (EC) No 1907/2006.

Date / Revised: 29.11.2023 Version: 3.0
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Product: **Na-Methylate Crystals**

(ID no. 30036694/SDS_GEN_FR/EN)

Date of print 11.07.2025

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Na-Methylate Crystals

Chemical name: sodium methylate crystals
CAS Number: 124-41-4

REACH registration number: 01-2119519241-51-0012, 01-2119519241-51-0001

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: industrial chemicals
Recommended use: initial product for chemical syntheses, process chemical

For the detailed identified uses of the product see appendix of the safety data sheet.

1.3. Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY

Contact address:
BASF France SAS
176, rue Montmartre
75002 PARIS
FRANCE

Telephone: +33 1 4964-5732
E-mail address: securite-produits.france@basf.com

1.4. Emergency telephone number

Tél.: 01 45 42 59 59 (APPEL D'URGENCE ORFILA)
Fax: 01 49 64 53 80 (heures de bureau)

International emergency number (Numéro d'urgence international):
contact speaking the language of the calling country (contact parlant la langue du pays d'appel)
Telephone: +49 180 2273-112

SECTION 2: Hazards Identification

2.1. Classification of the substance or mixture

According to Regulation (EC) No 1272/2008 [CLP]

Flam. Sol. 1	H228 Flammable solid.
Self-heat. 1	H251 Self-heating: may catch fire.
Acute Tox. 4 (oral)	H302 Harmful if swallowed.
Eye Dam./Irrit. 1	H318 Causes serious eye damage.
Skin Corr./Irrit. 1B	H314 Causes severe skin burns and eye damage.

According to BASF current knowledge and application of the criteria given in Annex I of Regulation (EC) No. 1272/2008, the following classification exceeding the classification given in Regulation (EC) No 1272/2008, Annex VI, Table 3.1 is required.

Flam. Sol. 1
Self-heat. 1
Acute Tox. 4 (oral)
Skin Corr./Irrit. 1A
Eye Dam./Irrit. 1

For the classifications not written out in full in this section the full text can be found in section 16.

2.2. Label elements

According to Regulation (EC) No 1272/2008 [CLP]

Pictogram:

Signal Word:

Danger

Hazard Statement:

H228	Flammable solid.
H251	Self-heating: may catch fire.
H302	Harmful if swallowed.
H314	Causes severe skin burns and eye damage.

Precautionary Statements (Prevention):

P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P280	Wear protective gloves, protective clothing and eye protection or face protection.

Precautionary Statements (Response):

P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310	Immediately call a POISON CENTER or physician.

Precautionary Statements (Storage):

P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

According to Regulation (EC) No 1272/2008 [CLP]

Labeling of special preparations (GHS):

EUH014: Reacts violently with water.

According to Regulation (EC) No 1272/2008 [CLP]EUH071: Corrosive to the respiratory tract.

Hazard determining component(s) for labelling: sodium methanolate

2.3. Other hazards

According to Regulation (EC) No 1272/2008 [CLP]

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.
When finely distributed, self-ignition is possible.
The product does not contain a substance above legal limits fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.
Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

SECTION 3: Composition/Information on Ingredients

3.1. Substances

Chemical nature

sodium methanolate

Content (W/W): 100 %	Flam. Sol. 1
CAS Number: 124-41-4	Self-heat. 1
EC-Number: 204-699-5	Acute Tox. 4 (oral)
INDEX-Number: 603-040-00-2	Skin Corr./Irrit. 1B
	Eye Dam./Irrit. 1
	H228, H251, H302, H314

EUH014
, EUH071

Differing classification according to current knowledge and the criteria given in Annex I of Regulation (EC) No. 1272/2008

Flam. Sol. 1
Self-heat. 1
Acute Tox. 4 (oral)
Skin Corr./Irrit. 1A
Eye Dam./Irrit. 1
EUH014, EUH071

For the classifications not written out in full in this section, including the hazard classes and the hazard statements, the full text is listed in section 16.

3.2. Mixtures

Not applicable

SECTION 4: First-Aid Measures

4.1. Description of first aid measures

First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention. Immediately administer a corticosteroid from a controlled/metered dose inhaler.

On skin contact:
Immediately wash thoroughly with plenty of water, apply sterile dressings, consult a skin specialist.

On contact with eyes:
Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200 - 300 ml water, do not induce vomiting, seek medical attention. Administer 50 ml of pure ethanol in a drinkable concentration. Seek medical attention.

4.2. Most important symptoms and effects, both acute and delayed

Symptoms: skin corrosion, Eye irritation, Further symptoms are possible

Hazards: No hazard is expected under intended use and appropriate handling.

4.3. Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

SECTION 5: Fire-Fighting Measures

5.1. Extinguishing media

Suitable extinguishing media:
dry powder, Dry sand, alcohol-resistant foam

Unsuitable extinguishing media for safety reasons:
water, carbon dioxide

5.2. Special hazards arising from the substance or mixture

Advice: Reacts violently with water. See SDS section 7 - Handling and storage.

Endangering substances: sodium oxides, organic vapours, corrosive gases/vapours, carbon oxides
Advice: Evolution of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

5.3. Advice for fire-fighters

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

SECTION 6: Accidental Release Measures

6.1. Personal precautions, protective equipment and emergency procedures

Avoid contact with the skin, eyes and clothing. Use breathing apparatus if exposed to vapours/dust/aerosol. Use personal protective clothing.

6.2. Environmental precautions

Do not allow to enter soil, waterways or waste water channels.

6.3. Methods and material for containment and cleaning up

For small amounts: Sweep/shovel up. Correctly dispose of recovered product immediately.
For large amounts: Sweep/shovel up. Correctly dispose of recovered product immediately.

6.4. Reference to other sections

Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

SECTION 7: Handling and Storage

7.1. Precautions for safe handling

Ensure thorough ventilation of stores and work areas. Breathing must be protected when large quantities are decanted without local exhaust ventilation. Protect against moisture. Protect from air. Protect from direct sunlight. Handle in protective atmosphere.

Protection against fire and explosion:
Take precautionary measures against static discharges. Sources of ignition should be kept well clear. Fire extinguishers should be kept handy. Avoid dust formation.

7.2. Conditions for safe storage, including any incompatibilities

Segregate from acids and acid forming substances.

Suitable materials for containers: Low density polyethylene (LDPE), Stainless steel 1.4301 (V2), Stainless steel 1.4401, glass, High density polyethylene (HDPE), Carbon steel (Iron), Stainless steel 1.4541, Stainless steel 1.4571, Alkyd resin lacquer 441
Further information on storage conditions: Keep container tightly closed in a cool, well-ventilated place. Keep under nitrogen.

7.3. Specific end use(s)

See exposure scenario(s) in the attachment to this safety data sheet.

SECTION 8: Exposure Controls/Personal Protection

8.1. Control parameters

Components with occupational exposure limits

The mentioned substance is result of gradual decomposition under influence of atmospheric humidity.

67-56-1: methanol

Skin Designation (OEL (EU))
The substance can be absorbed through the skin.
TWA value 260 mg/m3 ; 200 ppm (OEL (EU))
indicative

VLE (FR) 1.300 mg/m3 ; 1.000 ppm (OEL-IRNS (FR))
Ceiling limit value/factor: 15 min
Indicative limit
1310-73-2: sodium hydroxide

PNEC

freshwater: 154 mg/l

marine water: 15,4 mg/l

intermittent release: 1540 mg/l

sediment (freshwater): 570,4 mg/kg

sediment (marine water): 57,04 mg/kg

soil: 23,5 mg/kg

STP: 100 mg/l

oral (secondary poisoning):
No PNEC oral derived, as accumulation in organisms is not to be expected.

DNEL

No DNELs have been derived.

8.2. Exposure controls

Personal protective equipment

Respiratory protection:
Breathing protection if breathable aerosols/dust are formed. Particle filter with medium efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P2 or FFP2)

Hand protection:
Use gauntlets.
Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
butyl rubber (butyl) - 0.7 mm coating thickness
fluoroelastomer (FKM) - 0.7 mm coating thickness
Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing. Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:
Tightly fitting safety goggles (cage goggles) (e.g. EN 166) and face shield.

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures

Avoid contact with the skin, eyes and clothing. Do not breathe dust. Handle in accordance with good industrial hygiene and safety practice. Avoid inhalation of dusts.

SECTION 9: Physical and Chemical Properties

9.1. Information on basic physical and chemical properties

State of matter:	solid	
Form:	powder, crystalline	
Colour:	colourless	
Odour:	odourless	
Odour threshold:	Not determined due to potential health hazard by inhalation.	
melting point (decomposition):	> 350 °C The substance / product decomposes therefore not determined.	(Directive 92/69/EEC, A.1)
Boiling point:	> 350 °C (1.013,25 hPa) The substance / product decomposes therefore not determined.	(Directive 92/69/EEC, A.2)
Flammability:	Flammable solid., Highly flammable.	(Directive 84/449/EEC, A.10)
Lower explosion limit:	For solids not relevant for classification and labelling.	
Upper explosion limit:	For solids not relevant for classification and labelling.	
Flash point:	not applicable, Study scientifically not justified.	
Self-ignition temperature:	Temperature: > 25 - < 50 °C Pressure: 1.013 hPa	Test type: Self-ignition at high temperatures. (Method: Directive 92/69/EEC, A.16)
	No self ignition was observed up to the specified temperature.	
Thermal decomposition:	> 280 °C (DTA) Thermal decomposition above the indicated temperature is possible. The indicated value is for inert gas atmosphere.	

> 50 °C (VDI 2263, sheet 1, 1.4.1)
Risk of spontaneous ignition when exposed to air.

pH value: 12,8
(10 g/l, 20 °C)
Literature data.

Viscosity, kinematic:
Study technically not feasible.

Viscosity, dynamic:
Study technically not feasible.

Study technically not feasible.

Solubility in water: Study scientifically not justified.

Solubility (qualitative) solvent(s): alcohols
soluble

Partitioning coefficient n-octanol/water (log Kow): -0,72 (calculated)
(25 °C; pH value: < 13)

Information on: methanol
Partitioning coefficient n-octanol/water (log Kow): -0,77 (measured)
(20 °C)
Literature data.

Vapour pressure: < 0,000001 hPa (calculated)
(25 °C)

Relative density:
No data available.

Density: 1,3 g/cm3
(20 °C)
Literature data.

Relative vapour density (air):
The product is a non-volatile solid.

Particle characteristics

Particle size distribution: 3,6 µm (D10, ISO 13320-1)
135,7 µm (D90, ISO 13320-1)
69,9 µm (D50, ISO 13320-1)

9.2. Other information

Information with regard to physical hazard classes

Explosives

Explosion hazard: Based on the chemical structure there is no indication of explosive properties.

Impact sensitivity:
Based on the chemical structure there is no shock-sensitivity.

Oxidizing properties

Fire promoting properties: not fire-propagating
Study scientifically not justified.

Flammable solids

Burning rate: approx. 15 mm/s

Pyrophoric properties

Self-ignition temperature: Test type: Spontaneous self-ignition at room-temperature.

not self-igniting

Self-heating substances and mixtures

Self heating ability: It is a substance capable of spontaneous heating.

Substances and mixtures, which emit flammable gases in contact with water

Formation of flammable gases: (UN Test N.5 (contact with water))

Forms no flammable gases in the presence of water.

Corrosion to metals

Corrodes metals in the presence of water or moisture.

Other safety characteristics

Bulk density:	500 - 600 kg/m3 (< 40 °C)	(DIN 53466)
pKA:	15,17 (20 °C)	(calculated)
Hygroscopy:	hygroscopic	
Adsorption/water - soil:	KOC: 1 The product has not been tested. The statement has been derived from the properties of the hydrolysis products.	(calculated)
Surface tension:	Based on chemical structure, surface activity is not to be expected.	
Evaporation rate:	The product is a non-volatile solid.	

SECTION 10: Stability and Reactivity

10.1. Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals:	Corrodes metals in the presence of water or moisture.	
Formation of flammable gases:	Remarks:	Forms no flammable gases in the presence of water.
	Method:	Manual of tests and criteria. Test N.5 (United Nations Recommendations on the Transport of Dangerous Goods).

10.2. Chemical stability

The product is stable if stored and handled as prescribed/indicated.

10.3. Possibility of hazardous reactions

Exothermic reaction. Reacts with water and acids. Reacts with substances which contain active hydrogen. Self heating possible in the presence of air. Accumulation of fine dust may entail the risk of a dust explosion in the presence of air.

10.4. Conditions to avoid

Avoid all sources of ignition: heat, sparks, open flame. Avoid moisture. Avoid electro-static charge. Avoid heat.

10.5. Incompatible materials

Substances to avoid:
water, acids

10.6. Hazardous decomposition products

Hazardous decomposition products:
methanol, sodium hydroxide

SECTION 11: Toxicological Information

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Assessment of acute toxicity:

Of moderate toxicity after single ingestion. The toxicity of the product is based on its corrosivity.

Experimental/calculated data:

(by inhalation):Study does not need to be conducted.
LD50 rat (dermal): > 2.000 mg/kg (BASF-Test)
No mortality was observed. An aqueous solution was tested.

Information on: methanol
Assessment of acute toxicity:

Of high toxicity after single ingestion. Of high toxicity after short-term inhalation. Of high toxicity after short-term skin contact.

Irritation

Assessment of irritating effects:

Corrosive! Damages skin and eyes.

Experimental/calculated data:

Skin corrosion/irritation

rabbit: Corrosive. (similar to OECD guideline 404)

Serious eye damage/irritation

rabbit: irreversible damage (BASF-Test)

Respiratory/Skin sensitization

Assessment of sensitization:

As the substance is corrosive, conducting sensitization studies is not feasible. The chemical structure does not suggest a sensitizing effect.

Germ cell mutagenicity

Assessment of mutagenicity:

The substance was not mutagenic in bacteria. The substance was not mutagenic in mammalian cell culture. The substance was not mutagenic in a test with mammals. The product has not been fully tested. The statements have been derived in parts from products of a similar structure or composition.

Carcinogenicity

Assessment of carcinogenicity:
Study does not need to be conducted. The chemical structure does not suggest a specific alert for such an effect.

Reproductive toxicity

Assessment of reproduction toxicity:
Study does not need to be conducted. The chemical structure does not suggest a specific alert for such an effect.

Developmental toxicity

Assessment of teratogenicity:

Study does not need to be conducted. The chemical structure does not suggest a specific alert for such an effect.

Specific target organ toxicity (single exposure)

No data available.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

Study does not need to be conducted. The chemical structure does not suggest a specific alert of toxicity on target organs after repeated exposure.

Information on: methanol
Assessment of repeated dose toxicity:
The substance may cause blindness after repeated ingestion. The substance may cause blindness after repeated inhalation.

Aspiration hazard

Harmful if swallowed.

Interactive effects

No data available.

11.2. Information on other hazards

Endocrine disrupting properties

The substance is not identified to have endocrine disrupting properties according to Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 nor is included in the Candidate List of substances of very high concern according to EU REACh Article 59 for having endocrine disrupting properties.

SECTION 12: Ecological Information

12.1. Toxicity

Assessment of aquatic toxicity:
There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.
The product has not been tested. The statement has been derived from the properties of the hydrolysis products. The product gives rise to pH shifts.

Toxicity to fish:
LC50 (96 h) 15.400 mg/l, Lepomis macrochirus (Fish test acute, Flow through.)
Literature data. The product has not been tested. The statement has been derived from the properties of the hydrolysis products.

Aquatic invertebrates:
EC50 (96 h) 18.260 mg/l, Daphnia magna (DIN 38412 Part 11, semistatic)

Literature data. The product has not been tested. The statement has been derived from the properties of the hydrolysis products.

Aquatic plants:
EC50 (96 h) approx. 22.000 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201, static)
Literature data. The product has not been tested. The statement has been derived from the properties of the hydrolysis products.

Microorganisms/Effect on activated sludge:
EC50 (3 h) > 1.000 mg/l, activated sludge (OECD Guideline 209, static)
Literature data. The product has not been tested. The statement has been derived from the properties of the hydrolysis products.

Chronic toxicity to fish:
No observed effect concentration (200 h) 7.900 mg/l, Oryzias latipes (static)
The product has not been tested. The statement has been derived from the properties of the hydrolysis products.

No observed effect concentration (30 d) 450 mg/l, Pimephales promelas (calculated)
The product has not been tested. The statement has been derived from the properties of the hydrolysis products.

Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d) 208 mg/l, Daphnia magna (calculated)
The product has not been tested. The statement has been derived from the properties of the hydrolysis products.

Information on:sodium hydroxide
Assessment of aquatic toxicity:
Depending on local conditions and existing concentrations, disturbances in the biodegradation process of activated sludge are possible. There is a high probability that the product is not acutely harmful to aquatic organisms.
The effect strongly depends on the pH-value. The data refers to the dissociated form of the substance.

Information on:methanol
Assessment of aquatic toxicity:
There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.
There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Information on:sodium hydroxide
Toxicity to fish:
LC50 (96 h) 125 mg/l, Gambusia affinis (other, static)

The product will cause changes in the pH value of the test system. The result refers to an unneutralized sample. Literature data.

Information on:methanol
Toxicity to fish:
LC50 (96 h) 15.400 mg/l, Lepomis macrochirus (other, Flow through.)

Information on:sodium hydroxide
Aquatic invertebrates:
EC50 (48 h) 40,4 mg/l, Ceriodaphnia sp. (other, static)
Literature data.

Information on:methanol
Aquatic invertebrates:
EC50 (48 h) 18.260 mg/l, Daphnia magna (OECD Guideline 202, part 1, semistatic)

Information on:methanol
Aquatic plants:
EC50 (96 h) approx. 22.000 mg/l (growth rate), Selenastrum capricornutum (OECD Guideline 201, static)

Information on:methanol
Microorganisms/Effect on activated sludge:
EC50 (3 h) > 1.000 mg/l, (OECD Guideline 209, aquatic)

EC50 (24 h) 880 mg/l, Nitrosomonas sp. (Inhibition of nitrification, aquatic)

Assessment of terrestrial toxicity:
No toxic effects have been observed in terrestrial studies.

Soil living organisms:
No observed effect concentration (63 d) 10.000 mg/kg, Eisenia foetida (OECD Guideline 222, artificial soil)
The details of the toxic effect relate to the nominal concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Terrestrial plants:
EC50 41.000 mg/l, Lactuca sativa
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

No observed effect concentration 1.555 mg/kg, terrestrial plants
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Other terrestrial non-mammals:
No data available.

12.2. Persistence and degradability

Assessment biodegradation and elimination (H2O):
Readily biodegradable (according to OECD criteria).

Elimination information:
90 - 100 % BOD of the ThOD (20 d) (aerobic, activated sludge, domestic, non-adapted)
Literature data. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on:methanol
Assessment biodegradation and elimination (H2O):
Readily biodegradable (according to OECD criteria).

Information on:methanol
Elimination information:
95 % BOD of the ThOD (20 d) (OECD 301D; 92/69/EWG, C.4-E) (aerobic, activated sludge, domestic, non-adapted) Readily biodegradable (according to OECD criteria).

Assessment of stability in water:
In contact with water the substance will hydrolyse rapidly.
Study technically not feasible.

12.3. Bioaccumulative potential

Assessment bioaccumulation potential:
Does not significantly accumulate in organisms.

Bioaccumulation potential:

Bioconcentration factor(BCF): 4,5 (72 h), Cyprinus carpio (measured)
The product has not been tested. The statement has been derived from the properties of the hydrolysis products.

Information on:sodium hydroxide
Assessment bioaccumulation potential:
Accumulation in organisms is not to be expected.

Information on:methanol
Assessment bioaccumulation potential:
Significant accumulation in organisms is not to be expected.

12.4. Mobility in soil

Assessment transport between environmental compartments:
Volatility: The substance will not evaporate into the atmosphere from the water surface.
Adsorption in soil: Adsorption to solid soil phase is not expected.

12.5. Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not fulfill the criteria for PBT (Persistent/bioaccumulative/toxic) and vPvB (very persistent/very bioaccumulative). Self classification

12.6. Endocrine disrupting properties

The substance is not identified to have endocrine disrupting properties according to Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 nor is included in the Candidate List of substances of very high concern according to EU REACh Article 59 for having endocrine disrupting properties.

12.7. Other adverse effects

The substance is not listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

12.8. Additional information

Adsorbable organically-bound halogen (AOX):
This product contains no organically-bound halogen.

Other ecotoxicological advice:
Do not release untreated into natural waters. Due to the pH-value of the product, neutralization is generally required before discharging sewage into treatment plants. Inhibition of degradation activity in activated sludge is not to be anticipated during correct introduction of low concentrations. The local regulations on waste-water treatment must be followed.

SECTION 13: Disposal Considerations

13.1. Waste treatment methods

Hydrolyze product with excess of water under usage of the personal protection equipment and dispose of in accordance with local authority regulations.
Obtain the consent of pollution control authorities before discharging to wastewater treatment plants.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible; then it can be passed on for recycling after being thoroughly cleaned.

SECTION 14: Transport Information

Land transport

ADR

UN number or ID number:	UN1431
UN proper shipping name:	SODIUM METHYLATE
Transport hazard class(es):	4.2, 8
Packing group:	II
Environmental hazards:	no
Special precautions for user:	Tunnel code: D/E

RID

UN number or ID number:	UN1431
UN proper shipping name:	SODIUM METHYLATE
Transport hazard class(es):	4.2, 8
Packing group:	II
Environmental hazards:	no
Special precautions for user:	

Inland waterway transport
ADN

UN number or ID number:	UN1431
UN proper shipping name:	SODIUM METHYLATE
Transport hazard class(es):	4.2, 8
Packing group:	II
Environmental hazards:	no
Special precautions for user:	

Transport in inland waterway vessel
Not evaluated

Sea transport

IMDG

UN number or ID number:	UN 1431
UN proper shipping name:	SODIUM METHYLATE
Transport hazard class(es):	4.2, 8
Packing group:	II
Environmental hazards:	no Marine pollutant: NO
Special precautions for user:	EmS: F-A; S-L

Air transport

IATA/ICAO

UN number or ID number:	UN 1431
UN proper shipping name:	SODIUM METHYLATE
Transport hazard class(es):	4.2, 8
Packing group:	II
Environmental hazards:	No Mark as dangerous for the environment is needed
Special precautions for user:	

14.1. UN number or ID number
See corresponding entries for “UN number or ID number” for the respective regulations in the tables above.

14.2. UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

14.3. Transport hazard class(es)
See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

14.4. Packing group
See corresponding entries for “Packing group” for the respective regulations in the tables above.

14.5. Environmental hazards
See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

14.6. Special precautions for user

See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

14.7. Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

Further information

Specific national features of transport regulations must be observed. They are to be found in the shipping documents.

SECTION 15: Regulatory Information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

Prohibitions, Restrictions and Authorizations

Annex XVII of Regulation (EC) No 1907/2006: Number on List: 40, 75, 69

Directive 2012/18/EU - Control of Major Accident Hazards involving dangerous substances (EU): List entry in regulation: O1

Storage class in France (Nomenclature ICPE): 1450, 4610
Occupational Disease Number in France (Code de la Sécurité Sociale, Art. L 461-1 à L 461-8): 84

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

15.2. Chemical Safety Assessment

Chemical Safety Assessment performed

SECTION 16: Other Information

Full text of the classifications, including the hazard classes and the hazard statements, if mentioned in section 2 or 3:

Flam. Sol.	Flammable solids
Self-heat.	Self-heating substances and mixtures
Acute Tox.	Acute toxicity
Eye Dam./Irrit.	Serious eye damage/eye irritation
Skin Corr./Irrit.	Skin corrosion/irritation

H228	Flammable solid.
H251	Self-heating: may catch fire.
H302	Harmful if swallowed.
H314	Causes severe skin burns and eye damage.

Abbreviations
ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road. ADN = The European Agreement concerning the International Carriage of Dangerous Goods by Inland waterways. ATE = Acute Toxicity Estimates. CAO = Cargo Aircraft Only. CAS = Chemical Abstract Service. CLP = Classification, Labelling and Packaging of substances and mixtures. DIN = German national organization for standardization. DNEL = Derived No Effect Level. EC50 = Effective concentration median for 50% of the population. EC = European Community. EN = European Standards. IARC = International Agency for Research on Cancer. IATA = International Air Transport Association. IBC-Code = Intermediate Bulk Container code. IMDG = International Maritime Dangerous Goods Code. ISO = International Organization for Standardization. STEL = Short-Term Exposure Limit. LC50 = Lethal concentration median for 50% of the population. LD50 = Lethal dose median for 50% of the population. TLV = Threshold Limit Value. MARPOL = The International Convention for the Prevention of Pollution from Ships. NEN = Dutch Norm. NOEC = No Observed Effect Concentration. OEL = Occupational Exposure Limit. OECD = Organization for Economic Cooperation and Development. PBT = Persistent, Bioaccumulative and Toxic. PNEC = Predicted No Effect Level. PPM = Parts per million. RID = The European Agreement concerning the International Carriage of Dangerous Goods by Rail. TWA = Time Weight Average. UN-number = UN number at transport. vPvB = very Persistent and very Bioaccumulative.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Annex: Exposure Scenarios

Index

1. Distribution of substance
ERC2; PROC1, PROC5, PROC8a, PROC8b, PROC9, PROC15

2. Use as an intermediate
SU4, SU8, SU9, SU11, SU12; ERC6a; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC8b, PROC9

3. Use in chemical synthesis
SU4, SU8, SU9, SU11, SU12; ERC6b; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC8b, PROC9, PROC15

4. Use as laboratory reagent/agent
ERC8b; PROC15

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1. Short title of exposure scenario

Distribution of substance
ERC2; PROC1, PROC5, PROC8a, PROC8b, PROC9, PROC15

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ERC2: Formulation into mixture As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed.
Operational conditions	

Contributing exposure scenario	
Use descriptors covered	PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. Use domain: industrial
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	
Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented.	

Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or maintenance. Minimise number of staff exposed. Ensure that near miss events are documented. Ensure minimization of manual phases	
Containment as appropriate Ensure containment of the emmision source and provide extract ventilation to points where emission occur Handle substance within closed system. Ensure samples are obtained under containment or extract ventilation.	
Personal measures have to be applied in case of potential exposure only., Wear suitable coveralls to prevent exposure to the skin., Avoid inhalation of the product., Wear suitable chemically resistant gloves with sleeves., Wear suitable face shield, Wear suitable working clothes., Change gloves, if duration of activity exceeds break through time	
Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls	

to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

Contributing exposure scenario	
Use descriptors covered	PROC5: Mixing or blending in batch processes Use domain: industrial
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	
Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or maintenance. Minimise number of staff exposed. Ensure that near miss events are documented. Ensure minimization of manual phases	
Containment as appropriate Ensure containment of the emmision source and provide extract ventilation to points where emission occur Handle substance within closed system. Ensure samples are obtained under containment or extract ventilation.	
Personal measures have to be applied in case of potential exposure only., Wear suitable coveralls to prevent exposure to the skin., Avoid inhalation of the product., Wear	

suitable chemically resistant gloves with sleeves., Wear suitable face shield, Wear suitable working clothes., Change gloves, if duration of activity exceeds break through time	
Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Use domain: industrial
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	
Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid	

splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or maintenance. Minimise number of staff exposed. Ensure that near miss events are documented. Ensure minimization of manual phases	
Containment as appropriate Ensure containment of the emmision source and provide extract ventilation to points where emission occur Handle substance within closed system. Ensure samples are obtained under containment or extract ventilation.	
Personal measures have to be applied in case of potential exposure only., Wear suitable coveralls to prevent exposure to the skin., Avoid inhalation of the product., Wear suitable chemically resistant gloves with sleeves., Wear suitable face shield, Wear suitable working clothes., Change gloves, if duration of activity exceeds break through time	
Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Use domain: industrial
Operational conditions	

Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	
Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or maintenance. Minimise number of staff exposed. Ensure that near miss events are documented. Ensure minimization of manual phases	
Containment as appropriate Ensure containment of the emmision source and provide extract ventilation to points where emission occur Handle substance within closed system. Ensure samples are obtained under containment or extract ventilation.	
Personal measures have to be applied in case of potential exposure only., Wear suitable coveralls to prevent exposure to the skin., Avoid inhalation of the product., Wear suitable chemically resistant gloves with sleeves., Wear suitable face shield, Wear suitable working clothes., Change gloves, if duration of activity exceeds break through time	
Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict	

access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Use domain: industrial
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	
Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or maintenance. Minimise number of staff exposed. Ensure that near miss events are documented. Ensure minimization of manual phases	
Containment as appropriate Ensure containment of the emmision source and provide extract ventilation to points where emission occur Handle	

substance within closed system. Ensure samples are obtained under containment or extract ventilation.	
Personal measures have to be applied in case of potential exposure only., Wear suitable coveralls to prevent exposure to the skin., Avoid inhalation of the product., Wear suitable chemically resistant gloves with sleeves., Wear suitable face shield, Wear suitable working clothes., Change gloves, if duration of activity exceeds break through time	
Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

Contributing exposure scenario	
Use descriptors covered	PROC15: Use a laboratory reagent. Use domain: industrial
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	
Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and	

maintenance of equipment and machines. Restrict access to authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or maintenance. Minimise number of staff exposed. Ensure that near miss events are documented. Ensure minimization of manual phases	
Containment as appropriate Ensure containment of the emmision source and provide extract ventilation to points where emission occur Handle substance within closed system. Ensure samples are obtained under containment or extract ventilation.	
Personal measures have to be applied in case of potential exposure only., Wear suitable coveralls to prevent exposure to the skin., Avoid inhalation of the product., Wear suitable chemically resistant gloves with sleeves., Wear suitable face shield, Wear suitable working clothes., Change gloves, if duration of activity exceeds break through time	
Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

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2. Short title of exposure scenario

Use as an intermediate
SU4, SU8, SU9, SU11, SU12; ERC6a; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC8b, PROC9

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ERC6a: Use of intermediate As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed.
Operational conditions	

Contributing exposure scenario	
Use descriptors covered	PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. Use domain: industrial
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	
Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or maintenance. Minimise number of staff exposed. Ensure that near miss	

events are documented. Ensure minimization of manual phases	
Containment as appropriate Ensure containment of the emmision source and provide extract ventilation to points where emission occur Handle substance within closed system. Ensure samples are obtained under containment or extract ventilation.	
Personal measures have to be applied in case of potential exposure only., Wear suitable coveralls to prevent exposure to the skin., Avoid inhalation of the product., Wear suitable chemically resistant gloves with sleeves., Wear suitable face shield, Wear suitable working clothes., Change gloves, if duration of activity exceeds break through time	
Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Use domain: industrial
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	

Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or maintenance. Minimise number of staff exposed. Ensure that near miss events are documented. Ensure minimization of manual phases	
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Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear	

suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Use domain: industrial
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	
Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or maintenance. Minimise number of staff exposed. Ensure that near miss events are documented. Ensure minimization of manual phases	
Containment as appropriate Ensure containment of the emmision source and provide extract ventilation to points where emission occur Handle substance within closed system. Ensure samples are obtained under	

containment or extract ventilation.	
Personal measures have to be applied in case of potential exposure only., Wear suitable coveralls to prevent exposure to the skin., Avoid inhalation of the product., Wear suitable chemically resistant gloves with sleeves., Wear suitable face shield, Wear suitable working clothes., Change gloves, if duration of activity exceeds break through time	
Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

Contributing exposure scenario	
Use descriptors covered	PROC4: Chemical production where opportunity for exposure arises Use domain: industrial
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	
Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and	

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Personal measures have to be applied in case of potential exposure only., Wear suitable coveralls to prevent exposure to the skin., Avoid inhalation of the product., Wear suitable chemically resistant gloves with sleeves., Wear suitable face shield, Wear suitable working clothes., Change gloves, if duration of activity exceeds break through time	
Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Use domain: industrial
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	
Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or maintenance. Minimise number of staff exposed. Ensure that near miss events are documented. Ensure minimization of manual phases	
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Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Use domain: industrial
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	
Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or maintenance. Minimise number of staff exposed. Ensure that near miss	

events are documented. Ensure minimization of manual phases	
Containment as appropriate Ensure containment of the emmision source and provide extract ventilation to points where emission occur Handle substance within closed system. Ensure samples are obtained under containment or extract ventilation.	
Personal measures have to be applied in case of potential exposure only., Wear suitable coveralls to prevent exposure to the skin., Avoid inhalation of the product., Wear suitable chemically resistant gloves with sleeves., Wear suitable face shield, Wear suitable working clothes., Change gloves, if duration of activity exceeds break through time	
Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Use domain: industrial
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	
Provide specific employee training to	

prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or maintenance. Minimise number of staff exposed. Ensure that near miss events are documented. Ensure minimization of manual phases	
Containment as appropriate Ensure containment of the emmision source and provide extract ventilation to points where emission occur Handle substance within closed system. Ensure samples are obtained under containment or extract ventilation.	
Personal measures have to be applied in case of potential exposure only., Wear suitable coveralls to prevent exposure to the skin., Avoid inhalation of the product., Wear suitable chemically resistant gloves with sleeves., Wear suitable face shield, Wear suitable working clothes., Change gloves, if duration of activity exceeds break through time	
Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear suitable working clothes., Wear	

suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

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3. Short title of exposure scenario

Use in chemical synthesis
SU4, SU8, SU9, SU11, SU12; ERC6b; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC8b, PROC9, PROC15

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ERC6b: Use of reactive processing aid at industrial site (no inclusion into or onto article) As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed.
Operational conditions	

Contributing exposure scenario	
Use descriptors covered	PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. Use domain: industrial
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	
Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and machines. Restrict access to	

authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or maintenance. Minimise number of staff exposed. Ensure that near miss events are documented. Ensure minimization of manual phases	
Containment as appropriate Ensure containment of the emmision source and provide extract ventilation to points where emission occur Handle substance within closed system. Ensure samples are obtained under containment or extract ventilation.	
Personal measures have to be applied in case of potential exposure only., Wear suitable coveralls to prevent exposure to the skin., Avoid inhalation of the product., Wear suitable chemically resistant gloves with sleeves., Wear suitable face shield, Wear suitable working clothes., Change gloves, if duration of activity exceeds break through time	
Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

Contributing exposure scenario

Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Use domain: industrial
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	
Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or maintenance. Minimise number of staff exposed. Ensure that near miss events are documented. Ensure minimization of manual phases	
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Personal measures have to be applied in case of potential exposure only., Wear suitable coveralls to prevent exposure to the skin., Avoid inhalation of the product., Wear suitable chemically resistant gloves with sleeves., Wear suitable face shield, Wear suitable working clothes., Change gloves, if duration of activity exceeds break through time	

Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Use domain: industrial
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	
Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or	

maintenance. Minimise number of staff exposed. Ensure that near miss events are documented. Ensure minimization of manual phases	
Containment as appropriate Ensure containment of the emmision source and provide extract ventilation to points where emission occur Handle substance within closed system. Ensure samples are obtained under containment or extract ventilation.	
Personal measures have to be applied in case of potential exposure only., Wear suitable coveralls to prevent exposure to the skin., Avoid inhalation of the product., Wear suitable chemically resistant gloves with sleeves., Wear suitable face shield, Wear suitable working clothes., Change gloves, if duration of activity exceeds break through time	
Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

Contributing exposure scenario	
Use descriptors covered	PROC4: Chemical production where opportunity for exposure arises Use domain: industrial
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa

Risk Management Measures	
Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or maintenance. Minimise number of staff exposed. Ensure that near miss events are documented. Ensure minimization of manual phases	
Containment as appropriate Ensure containment of the emmision source and provide extract ventilation to points where emission occur Handle substance within closed system. Ensure samples are obtained under containment or extract ventilation.	
Personal measures have to be applied in case of potential exposure only., Wear suitable coveralls to prevent exposure to the skin., Avoid inhalation of the product., Wear suitable chemically resistant gloves with sleeves., Wear suitable face shield, Wear suitable working clothes., Change gloves, if duration of activity exceeds break through time	
Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	

Wear suitable face shield, Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Use domain: industrial
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	
Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or maintenance. Minimise number of staff exposed. Ensure that near miss events are documented. Ensure minimization of manual phases	
Containment as appropriate Ensure containment of the emmision source and provide extract ventilation to points where emission occur Handle substance within closed system. Ensure samples are obtained under containment or extract ventilation.	

Personal measures have to be applied in case of potential exposure only., Wear suitable coveralls to prevent exposure to the skin., Avoid inhalation of the product., Wear suitable chemically resistant gloves with sleeves., Wear suitable face shield, Wear suitable working clothes., Change gloves, if duration of activity exceeds break through time	
Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Use domain: industrial
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	
Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and machines. Restrict access to	

authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or maintenance. Minimise number of staff exposed. Ensure that near miss events are documented. Ensure minimization of manual phases	
Containment as appropriate Ensure containment of the emmision source and provide extract ventilation to points where emission occur Handle substance within closed system. Ensure samples are obtained under containment or extract ventilation.	
Personal measures have to be applied in case of potential exposure only., Wear suitable coveralls to prevent exposure to the skin., Avoid inhalation of the product., Wear suitable chemically resistant gloves with sleeves., Wear suitable face shield, Wear suitable working clothes., Change gloves, if duration of activity exceeds break through time	
Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

Contributing exposure scenario

Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Use domain: industrial
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	
Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or maintenance. Minimise number of staff exposed. Ensure that near miss events are documented. Ensure minimization of manual phases	
Containment as appropriate Ensure containment of the emmision source and provide extract ventilation to points where emission occur Handle substance within closed system. Ensure samples are obtained under containment or extract ventilation.	
Personal measures have to be applied in case of potential exposure only., Wear suitable coveralls to prevent exposure to the skin., Avoid inhalation of the product., Wear suitable chemically resistant gloves with sleeves., Wear suitable face shield, Wear suitable working clothes., Change gloves, if duration of activity exceeds break through time	
Avoid contact with eyes.	

Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

Contributing exposure scenario	
Use descriptors covered	PROC15: Use a laboratory reagent. Use domain: industrial
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	
Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or maintenance. Minimise number of staff exposed. Ensure that near miss events are documented. Ensure minimization of manual phases	

Containment as appropriate Ensure containment of the emmision source and provide extract ventilation to points where emission occur Handle substance within closed system. Ensure samples are obtained under containment or extract ventilation.	
Personal measures have to be applied in case of potential exposure only., Wear suitable coveralls to prevent exposure to the skin., Avoid inhalation of the product., Wear suitable chemically resistant gloves with sleeves., Wear suitable face shield, Wear suitable working clothes., Change gloves, if duration of activity exceeds break through time	
Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

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4. Short title of exposure scenario

Use as laboratory reagent/agent
ERC8b; PROC15

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ERC8b: Widespread use of reactive processing aid (no inclusion into or onto article, indoor) As no environmental hazard was identified no

	environmental-related exposure assessment and risk characterization was performed.
Operational conditions	

Contributing exposure scenario	
Use descriptors covered	PROC15: Use a laboratory reagent. Use domain: professional
Operational conditions	
Physical state	Solid, low dustiness
Vapour pressure of the substance during use	0,000001 Pa
Risk Management Measures	
Provide specific employee training to prevent/minimize exposures. Ensure good work practices are implemented. Avoid contact with contaminated tools. Avoid skin contact. Clean up contamination as soon as they occur. Ensure that no inhalable dusts are generated. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Wash off any skin contamination immediately. Avoid splashing. Avoid manual contact with wet work pieces Equipment must be decontaminated prior to repair or maintenance. Minimise number of staff exposed. Ensure that near miss events are documented. Ensure minimization of manual phases	
Containment as appropriate Ensure containment of the emmision source and provide extract ventilation to points where emission occur Handle substance within closed system. Ensure samples are obtained under containment or extract ventilation.	
Personal measures have to be applied in case of potential exposure only., Wear suitable coveralls to prevent exposure to the skin., Avoid inhalation of the product., Wear suitable chemically resistant gloves	

with sleeves., Wear suitable face shield, Wear suitable working clothes., Change gloves, if duration of activity exceeds break through time	
Avoid contact with eyes.	
Use suitable eye protection.	
Avoid skin contact. Change clothes immediately after contamination. Provide specific employee training to prevent/minimize exposures. Restrict access to authorised persons. Wash off any skin contamination immediately.	
Wear suitable face shield, Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves., Wear suitable coveralls to prevent exposure to the skin.	
Restrict access to authorised persons.	
Wear suitable respiratory protection.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment

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